

RULE-BASED AI CHATBOT

1. Project Title

Rule-Based AI Chatbot using Python

2. Objective

To develop a simple rule-based chatbot that responds to predefined user commands such as greetings, date, time, weather information, and help instructions.

3. Problem Statement

Users often require quick information without accessing multiple applications. This project demonstrates how a chatbot can interact with users and provide predefined responses using Python.

4. Technologies Used

- Python
- Datetime Module
- Random Module

5. Features

- Greeting functionality
- Display current date
- Display current time
- Simulated weather prediction
- Help command
- Exit functionality
- Dictionary-based command handling

6. Algorithm

1. Start the chatbot.
2. Display welcome message.
3. Accept user input.
4. Convert input to lowercase.
5. Check if command exists in dictionary.
6. Execute corresponding function.
7. If command is "exit" or "bye", terminate the program.
8. Otherwise display error message.
9. Repeat until user exits.

7. Workflow

Input → Command Processing → Function Execution → Output Display

8. Code Explanation

The chatbot uses a Python dictionary to map commands to functions. This eliminates long if-else statements and improves scalability.

Functions:

- `hello()` → Greets user
- `show_date()` → Displays current date
- `show_time()` → Displays current time
- `show_weather()` → Displays random weather
- `show_help()` → Displays available commands

9. Sample Output

Command: hello

Output: Hello User! How can I help you today?

Command: date

Output: Current Date

Command: time

Output: Current Time

Command: weather

Output: Random Weather Condition

10. Applications

- Customer support systems
- Educational assistants
- Virtual assistants
- Interactive information systems

11. Conclusion

The Rule-Based AI Chatbot successfully demonstrates basic Natural Language Interaction using predefined commands. The project highlights the use of Python functions, dictionaries, loops, and user interaction concepts.